

Ch6. Hypothesis testing and Confidence Intervals

6.1 Hypothesis testing: General Hypothesis

Let $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$, Then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad \hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, (\mathbf{X}'\mathbf{X})^{-1}\sigma^2)$$

- We are now interested in testing the following hypothesis:

$$H_0 : \mathbf{K}'\boldsymbol{\beta} = \boldsymbol{\mu} \tag{6.1}$$

where $\mathbf{K}'$ is $q \times (p+1)$, and $\mathbf{K}'$ is assumed to be full row rank.

6.1 Hypothesis testing: General Hypothesis

- The test statistics

$$\begin{aligned} F(H) &= \frac{Q/q}{SSE/[n - r(\mathbf{X})]} = \frac{Q}{q\hat{\sigma}^2} \\ &\sim F_{[q, n-r(\mathbf{X})], \frac{1}{2\sigma^2}(\mathbf{K}'\boldsymbol{\beta} - \boldsymbol{\mu})'[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}(\mathbf{K}'\boldsymbol{\beta} - \boldsymbol{\mu})} \end{aligned}$$

Here $\hat{\sigma}^2 = SSE/(n - r(\mathbf{X}))$ which is the unbiased estimator of σ^2 .

- Under $H_0 : \mathbf{K}'\boldsymbol{\beta} = \boldsymbol{\mu}$, we have

$$F(\mathbf{H}) \sim F_{\{q, n-r(\mathbf{X})\}}.$$

6.1 Hypothesis testing: General Hypothesis

Note that

■

$$\mathbf{K}'\hat{\beta} - \mu \sim N[\mathbf{K}'\beta - \mu, \mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}\sigma^2]$$

■ $(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}$ is symmetric.

■ Let

$$Q = (\mathbf{K}'\hat{\beta} - \mu)'[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}(\mathbf{K}'\hat{\beta} - \mu)$$

(Q is sometimes denoted by SSH) then

$$\frac{Q}{\sigma^2} \sim \chi^2_{\{q, \frac{1}{2\sigma^2}(\mathbf{K}'\beta - \mu)'[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}(\mathbf{K}'\beta - \mu)\}}$$

■ Q and SSE are independent

6.1 Hypothesis testing: General Hypothesis

Consider the following model

$$\mathbf{y} \sim N(\mathbf{X}\mathbf{a}, \sigma^2 \mathbf{I})$$

under the constraint (the null hypothesis) $H_0 : \mathbf{K}'\mathbf{a} = \mu$.

- Denote the LS estimator of $\mathbf{a}$ by $\tilde{\mathbf{a}}$.
- To obtain $\tilde{\mathbf{a}}$ by minimizing

$$(\mathbf{y} - \mathbf{X}\mathbf{a})'(\mathbf{y} - \mathbf{X}\mathbf{a}) + 2\theta'(\mathbf{K}'\mathbf{a} - \mu)$$

with respect to $\mathbf{a}$ and θ . Note that 2θ is a vector of Lagrange multipliers. After the minimization,

$$\begin{aligned} \tilde{\mathbf{a}} &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y} - \mathbf{K}(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - \mu)) \\ &= \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}(\mathbf{K}'\hat{\beta} - \mu) \end{aligned} \quad (6.2)$$

- $\hat{\beta} - \tilde{\mathbf{a}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}(\mathbf{K}'\hat{\beta} - \mu)$;
- $\tilde{\mathbf{a}}$ is the BLUE for the model with the constraint.

6.1 Proof of that $\tilde{a}$ is the BLUE for the model with the constraint

Proof.

Let $\mathbf{L} = (\mathbf{y} - \mathbf{X}\mathbf{a})'(\mathbf{y} - \mathbf{X}\mathbf{a})$. For $\forall \mathbf{a}_0$ with constraint $\mathbf{K}'\mathbf{a}_0 = \boldsymbol{\mu}$, we have

$$\begin{aligned}\mathbf{L}_0 &= (\mathbf{y} - \mathbf{X}\mathbf{a}_0)'(\mathbf{y} - \mathbf{X}\mathbf{a}_0) \\ &= (\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}} + \mathbf{X}\tilde{\mathbf{a}} - \mathbf{X}\mathbf{a}_0)'(\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}} + \mathbf{X}\tilde{\mathbf{a}} - \mathbf{X}\mathbf{a}_0) \\ &= (\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}})'(\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}}) + (\mathbf{X}\tilde{\mathbf{a}} - \mathbf{X}\mathbf{a}_0)'(\mathbf{X}\tilde{\mathbf{a}} - \mathbf{X}\mathbf{a}_0) \\ &\quad + 2(\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}})'(\mathbf{X}\tilde{\mathbf{a}} - \mathbf{X}\mathbf{a}_0)\end{aligned}$$

6.1 Proof (continued)

But

$$\begin{aligned}
 & (\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}})'(\mathbf{X}\tilde{\mathbf{a}} - \mathbf{X}\mathbf{a}_0) \\
 = & (\mathbf{y}'\mathbf{X} - \tilde{\mathbf{a}}'\mathbf{X}'\mathbf{X})(\tilde{\mathbf{a}} - \mathbf{a}_0) \\
 = & [\mathbf{y}'\mathbf{X} - \hat{\beta}'(\mathbf{X}'\mathbf{X}) + (\mathbf{K}'\hat{\beta} \\
 & - \mu)'(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})](\tilde{\mathbf{a}} - \mathbf{a}_0) \\
 = & (\mathbf{K}'\hat{\beta} - \mu)'(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}\mathbf{K}'(\tilde{\mathbf{a}} - \mathbf{a}_0) = 0
 \end{aligned}$$

Because $(\mathbf{K}'\tilde{\mathbf{a}} = \mathbf{K}'\mathbf{a}_0 = \mu)$. Thus,

$$\mathbf{L}_0 = (\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}})'(\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}}) + (\tilde{\mathbf{a}} - \mathbf{a}_0)'\mathbf{X}'\mathbf{X}(\tilde{\mathbf{a}} - \mathbf{a}_0).$$

Hence,

$$\mathbf{a}_0 = \tilde{\mathbf{a}} \text{ minimize } \mathbf{L}_0.$$

6.1 Hypothesis testing: General Hypothesis

- Without the null hypothesis, $SSE = (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})$.
- Under the null hypothesis (reduced Model)

$$\begin{aligned}
 SSE_{H_0} &= (\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}})'(\mathbf{y} - \mathbf{X}\tilde{\mathbf{a}}) \\
 &= [\mathbf{y} - \mathbf{X}\hat{\beta} + \mathbf{X}\hat{\beta} - \mathbf{X}\tilde{\mathbf{a}}]'[\mathbf{y} - \mathbf{X}\hat{\beta} + \mathbf{X}\hat{\beta} - \mathbf{X}\tilde{\mathbf{a}}] \\
 &= [\mathbf{y} - \mathbf{X}\hat{\beta} + \mathbf{X}(\hat{\beta} - \tilde{\mathbf{a}})]'[\mathbf{y} - \mathbf{X}\hat{\beta} + \mathbf{X}(\hat{\beta} - \tilde{\mathbf{a}})] \\
 &= (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) + (\hat{\beta} - \tilde{\mathbf{a}})' \mathbf{X}' \mathbf{X} (\hat{\beta} - \tilde{\mathbf{a}})
 \end{aligned}$$

Since $((\hat{\beta} - \tilde{\mathbf{a}})' \mathbf{X}' (\mathbf{y} - \mathbf{X}\hat{\beta}) = 0)$

- From (6.2),

$$\begin{aligned}
 SSE_{H_0} &= SSE + (\mathbf{K}'\hat{\beta} - \mu)'[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}\mathbf{X}'\mathbf{K}[\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K}]^{-1}(\mathbf{K}\hat{\beta} - \mu) \\
 &= SSE + (\mathbf{K}'\hat{\beta} - \mu)'(\mathbf{K}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{K})^{-1}(\mathbf{K}'\hat{\beta} - \mu) \\
 &= SSE + Q \\
 &\geq SSE
 \end{aligned}$$

6.1 Special cases. (1) $H_0 : \beta = \beta_0$

In this case, $\mathbf{K}' = \mathbf{I}$, $q = p + 1$, $\mu = \beta_0$

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$$Q = (\hat{\beta} - \beta_0)' \mathbf{X}' \mathbf{X} (\hat{\beta} - \beta_0)$$

2 $F(H) = \frac{Q}{(p+1)\hat{\sigma}^2}$

3 Under the null hypothesis,

$$F(H) \sim F_{\{p+1, n-(p+1)\}}$$

4 $\tilde{\mathbf{a}} = \hat{\beta} - (\hat{\beta} - \mathbf{a}_0) = \beta_0$

6.1 Special cases. (2) $H_0 : \lambda' \beta = m$

In this case, $K' = \lambda'$, $q = 1$, $\mu = m$

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$$\begin{aligned} Q &= (\lambda' \hat{\beta} - m)' [\lambda' (\mathbf{X}' \mathbf{X})^{-1} \lambda]^{-1} (\lambda' \hat{\beta} - m) \\ &= (\lambda' \hat{\beta} - m)^2 / \lambda' (\mathbf{X}' \mathbf{X})^{-1} \lambda; \end{aligned}$$

2 $F(H) = \frac{Q}{\hat{\sigma}^2};$

3 Under the null hypothesis,

$$F(H) \sim F_{(1, n-r(\mathbf{X}))}$$

Note: $\sqrt{F(H)} = \frac{\sqrt{Q}}{\hat{\sigma}} \sim t_{n-r(\mathbf{X})};$

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$$\begin{aligned} \tilde{\mathbf{a}} &= \hat{\beta} - (\mathbf{X}' \mathbf{X})^{-1} \lambda [\lambda' (\mathbf{X}' \mathbf{X})^{-1} \lambda]^{-1} (\lambda' \hat{\beta} - \mu) \\ &= \hat{\beta} - \frac{(\lambda' \hat{\beta} - \mu)}{\lambda' (\mathbf{X}' \mathbf{X})^{-1} \lambda} (\mathbf{X}' \mathbf{X})^{-1} \lambda. \end{aligned}$$

Note: $\lambda' \hat{\beta} - \mu \sim N(\lambda' \beta - \mu, \lambda' (\mathbf{X}' \mathbf{X})^{-1} \lambda \sigma^2)$

6.1 Special cases. (3) $H_0 : \beta_2 = 0$

Here, $\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$.

6.1 Likelihood Ratio Test

Theorem: If $\mathbf{y}$ is $N_n(\mathbf{x}\mathbf{a}, \sigma^2\mathbf{I})$, where (rank of $\mathbf{x}$ is $p + 1$), the likelihood ratio for $H_0: \mathbf{a} = \mathbf{0}$ can be based on

$$F = \frac{\hat{\mathbf{a}}' \mathbf{x}' \mathbf{y} / (p + 1)}{(\mathbf{y}' \mathbf{y} - \hat{\mathbf{a}}' \mathbf{x}' \mathbf{y}) / (n - p - 1)}.$$

H_0 is rejected if $F > F_{\alpha, p+1, n-p-1}$.

6.2 Confidence intervals and prediction intervals

- Confidence region for β
- Confidence interval for β_j
- Confidence interval for $\mathbf{X}'\beta$
- Confidence interval for $E(y^*)$ given $\mathbf{x} = \mathbf{x}^*$
- Prediction interval for a future observation
- Confidence interval for σ^2
- Simultaneous intervals
 - Familywise confidence level
 - Bonferroni procedure
 - Scheffé procedure

6.2 Confidence region for β

Since

$$P\left[\frac{(\hat{\beta} - \beta)' \mathbf{X}' \mathbf{X} (\hat{\beta} - \beta)}{(p+1)\hat{\sigma}^2} \leq F_{\alpha, p+1, n-p-1}\right] = 1 - \alpha,$$

a $100(1 - \alpha)\%$ joint confidence region for $\beta_0, \beta_1, \dots, \beta_p$ is defined to consist of all vectors β that satisfy

$$(\hat{\beta} - \beta)' \mathbf{X}' \mathbf{X} (\hat{\beta} - \beta) \leq (p+1)\hat{\sigma}^2 F_{\alpha, p+1, n-p-1}.$$

6.2 Confidence interval for β_j

Since

$$P[-t_{\alpha/2, n-p-1} \leq \frac{\hat{\beta}_j - \beta_j}{S\{\hat{\beta}_j\}} \leq t_{\alpha/2, n-p-1}] = 1 - \alpha,$$

hence, a $100(1 - \alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{\alpha/2, n-p-1} S\{\hat{\beta}_j\}.$$

6.2 Confidence interval for $\lambda'\beta$

Note that

$$t = \frac{\lambda'\hat{\beta} - \lambda'\beta}{\hat{\sigma}\sqrt{\lambda'(\mathbf{X}'\mathbf{X})^{-1}\lambda}} \sim t_{n-p-1},$$

hence, a $100(1 - \alpha)\%$ confidence interval for $\lambda'\beta$ is

$$\lambda'\hat{\beta} \pm t_{\alpha/2, n-p-1} \hat{\sigma} \sqrt{\lambda'(\mathbf{X}'\mathbf{X})^{-1}\lambda}.$$

6.2 Confidence interval for $E(y^*)$ given $x = x^*$

Given that $x = x^*$,

$$E(y^*) = x^{*'}\beta,$$

$$\widehat{E(y^*)} = x^{*'}\hat{\beta},$$

and

$$\text{Var}(E(y^*) - \widehat{E(y^*)}) = [x^{*'}(X'X)^{-1}x^*]\sigma^2,$$

hence, a $100(1 - \alpha)\%$ confidence interval for $E(y^*)$ is

$$x^{*'}\hat{\beta} \pm t_{\alpha/2, n-p-1}\hat{\sigma}\sqrt{x^{*'}(X'X)^{-1}x^*}.$$

6.2 Prediction interval for a future observation

Given that $\mathbf{x} = \mathbf{x}^*$. Let $y^* = \mathbf{x}'\boldsymbol{\beta} + \epsilon^*$ be a future value of y when $\mathbf{x} = \mathbf{x}^*$ that needs to be predicted.

$$\hat{y}^* = \mathbf{x}^{*'}\hat{\boldsymbol{\beta}},$$

$$\text{Var}(y^* - \hat{y}^*) = [1 + \mathbf{x}^{*'}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}^*]\sigma^2,$$

hence, a $100(1 - \alpha)\%$ prediction interval for y^* is

$$\hat{y}^* \pm t_{\alpha/2, n-p-1} \hat{\sigma} \sqrt{1 + \mathbf{x}^{*'}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}^*}.$$

6.2 Confidence interval for σ^2

Note that $(n - p - 1)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{(n-p-1)}$. Therefore,

$$P[\chi^2_{1-\alpha/2, n-p-1} \leq \frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} \leq \chi^2_{\alpha/2, n-p-1}] = 1 - \alpha,$$

hence, a $100(1 - \alpha)\%$ confidence interval for σ^2 is

$$\left(\frac{(n-p-1)\hat{\sigma}^2}{\chi^2_{\alpha/2, n-p-1}}, \frac{(n-p-1)\hat{\sigma}^2}{\chi^2_{1-\alpha/2, n-p-1}} \right).$$

6.2 Simultaneous intervals

- Familywise confidence level: $1 - \alpha_f$ implies that we are $100(1 - \alpha_f)\%$ confident that every interval contains its respective parameter.
- Bonferroni confidence intervals
 - Individual confidence level $1 - \alpha_c$;
 - m intervals;
 - If we choose $\alpha_c = \alpha_f/m$, familywise confidence level $\geq 1 - \alpha_f$;
 - For m linear functions $\lambda'_1\beta, \lambda'_2\beta, \dots, \lambda'_m\beta$, for $i = 1, \dots, m$, the $100(1 - \alpha)\%$ Bonferroni confidence intervals are

$$\lambda'_i\hat{\beta} \pm t_{\alpha/2m, n-p-1} \hat{\sigma} \sqrt{\lambda'_i(\mathbf{X}'\mathbf{X})^{-1}\lambda_i}.$$

- Scheffé confidence intervals for all possible linear functions $\lambda'\beta$: the $100(1 - \alpha)\%$ conservative confidence interval for any and all $\lambda'\beta$ is

$$\lambda'\hat{\beta} \pm \hat{\sigma} \sqrt{(p+1)F_{\alpha, p+1, n-p-1} \lambda'(\mathbf{X}'\mathbf{X})^{-1}\lambda}$$